

PROJECT GUIDELINE: COSC 594 – FALL 2025

Project Description: Project aims to present and discuss a machine learning paper on the interplay between RL and GenAI. Students work in a team of two to present an assigned paper and summarize it in a joint report in L^AT_EX. The team assignment is based on the alphabetical order of last names, except for a singular team. The project (50% of the course) consists of two parts:

- (20% presentation). This is an in-class joint presentation of the assigned paper. Presentation slides are due at the beginning of the scheduled class.
- (30% report). A joint final report that summarizes the assigned paper, which is due the last day of class (**12/2, 11:59pm**). Late submission will not be accepted.

Presentation Schedule: The presentation dates and papers are assigned below.

Class Date: 11/3

Presenters: Sanskar Amgain, Tingwei Chen

Paper: Decoding-Time Language Model Alignment with Multiple Objectives, NeurIPS 2024

Class Date: 11/5

Presenters: Nolan Coffey, Ethan Head

Paper: Improving LLM General Preference Alignment via Optimistic Online Mirror Descent, NeurIPS 2025

Class Date: 11/7

Presenters: Robert King, Sheng Liu

Paper: Accelerating RL for LLM Reasoning with Optimal Advantage Regression, NeurIPS 2025

Class Date: 11/10

Presenters: Madhusree Mondal, Kapildev Neupane

Paper: DSPO: Direct Score Preference Optimization for Diffusion Model Alignment, ICLR 2025

Class Date: 11/12

Presenters: Faithful Mari Odoi, Cameron Rader

Paper: The Superposition of Diffusion Models Using the Itô Density Estimator, ICLR 2025

Class Date: 11/14

Presenters: Stephen Nathaniel Souther, Utham Suresh

Paper: Diffusion-based Reinforcement Learning via Q-weighted Variational Policy Optimization, NeurIPS 2024

Class Date: 11/17

Presenters: Qipan Xu, Zecheng Zhan

Paper: SafeDiffuser: Safe Planning with Diffusion Probabilistic Models, ICLR 2025

Class Date: 11/19

Presenters: Bryce Bible

Paper: Posterior Inference with Diffusion Models for High-dimensional Black-box Optimization, ICML 2025

Reading Guideline: The assigned papers are closely related to the topics covered in the lectures. Most papers are selected from top-tier ML/AI conferences such as NeurIPS, ICLR, and ICML, with a balanced emphasis on both theory and experiments. You should be able to identify whether your assigned paper addresses a RL or GenAI problem, and connect it to the topics we have covered in class. Specific guiding questions are provided below.

- What problem does the assigned paper aim to address, and what connection of it with the lecture?
- What are the main ideas, methods, contributions of the assigned paper?
- How does the authors' results differ from those in related work or similar work?
- What are the strengths and limitations of the assigned paper?
- After discussing with your teammate, what insights or differing perspectives emerged?

If you need further help understanding the paper, feel free to ask on CANVAS or schedule office hours.

Presentation Guideline: In-class presentation is designed to help you practice giving research talks, which are the most effective way to communicate and promote your research. To ensure a smooth presentation, please follow the instructions below.

Each presentation should be about 40 minutes long (including transition), followed by 10 minutes for Q&A. Each of you should present for approximately 20 minutes (you may take turns or split the talk by sections). I strongly recommend the following format for your presentation:

- (≈ 4 minutes). **Introduction:** What is the topic of the assigned paper, and why is it important? How is it related to the course? What are the main contributions? What will you present in the rest of the talk?
- (≈ 12 minutes). **Problem setup:** What is the problem (including notation)? Are there existing methods that can be applied? What are the main challenges, and why are they difficult to address (in short)?
- (≈ 20 minutes). **Main results:** What are the new formulations, algorithms, or theorems presented in the assigned paper? What are the key ideas or intuitions behind them? How do the authors use these to address the main challenges (in detail)? Are there effective experiments that support the main results? How are the methods or algorithms implemented and evaluated?
- (≈ 4 minutes). **Conclusion:** What are the main take-home messages? What open questions remain, and how might they connect to future research or course topics?

During preparation, below are some helpful points.

- For a 40-minutes talk, the total number of slides (excluding transition slides) should be around 30. Make sure your slides readable from the back of class, and use consistent mathematical notation and clear visual illustrations. Before class, confirm that your computer connects to the classroom projector properly.
- Be prepared to take questions during the presentation. Make sure you understand each question and respond concisely. It is perfectly fine if you don't have an answer, and in this case simply acknowledge it and continue your talk. For the audience, it is strongly recommended to ask questions at any point during the presentation; doing so will count toward class participation (10% of the course).
- You do not need to cover everything in the assigned paper. In most cases, you only have time to present one or two main results. Let the audience know which parts you will focus on before presenting the main results.
- Whether the paper is more theoretical or empirical, your talk does not need to follow that style exactly. It is often helpful to simplify notation when explaining complex math, draw diagrams for lengthy algorithms, and summarize key findings when discussing large experiments.

For a single-person team, the presentation time is reduced by half, and the same guideline applies.
If you have questions about your presentation, please post them on CANVAS or come to office hours.

Report Writing Guideline: The final joint report must be written using the L^AT_EX template of NeurIPS and be at least 6 pages long, excluding references and appendices. You can find the NeurIPS L^AT_EX format on this page (<https://neurips.cc/Conferences/2025/CallForPapers>). The report should summarize the main results of the assigned paper in your own words, and include the following components:

- **Abstract:** Summarize the entire report in a paragraph.
- **Introduction:** (1) Explain the motivation of the study in the assigned paper and how it relates to the course; (2) Describe the key challenges the authors aim to address; (3) Summarize the main contributions of the paper at a high level.
- **Problem setup:** Describe the problem formulation using appropriate notation. You do not need to include every detail, and focus on a few key aspects that fit within the page limit of the report.
- **Main results:** Present the new formulations, algorithms, or theorems. For new formulations, highlight what is novel and describe the key ideas or techniques. For theorems, provide a formal statement and explain, in your words, what the statement really means and what its implications are.
- **Derivations and proofs:** Present the derivations of the new formulations, algorithms, and theorems. Select a few important or inspiring derivations or proofs from the paper and reproduce them in your report. Again, you do not need to cover all derivations and proofs. Try to break them into clear parts or steps to aid understanding.
- **Experiments:** Select the most effective experiments that convincingly support the main results or demonstrate the performance of the new algorithms or methods. Clearly state the experimental setup, including the environment or dataset used.
- **Conclusion:** Summarize the report in a short paragraph, highlighting the main takeaways again. Additionally, provide a brief contribution statement in a separate paragraph, indicating which parts were written by each team member (it is fine if some parts were written jointly).
- **References and appendices:** List all references in your report, and include any supplementary materials in the appendices, if applicable.

Since the report is written jointly by two team members, ensure consistency in notation, terminology, and formatting throughout the entire report (these will be evaluated closely). You are welcome to set up an appointment to discuss the strengths and weaknesses of your draft.

For a single-person team, the number of pages is reduced by half, and the same guideline applies.

Bonus Point: Two optional tasks are considered as bonus (10% of the course).

- **Open questions:** Identify interesting and concrete open questions in the same research direction that are not mentioned in the assigned paper. Discuss potential approaches to address these questions and/or explain why they are challenging problems that may require new formalizations, designs, or algorithms beyond those presented in the paper.
- **More papers in the same direction:** Find and summarize additional papers on the same topic, highlighting what they study and how they relate to or differ from your assigned paper. You may use the reference list of your assigned paper, check which papers cite it on Google Scholar, or perform a general online search on the related topic.

For bonus point tasks, you may include additional pages after the Conclusion section and before the References. Please indicate the type of bonus task and which team member wrote it (it is allowed to be written jointly).

It is sufficient for one team member to submit the final report through CANVAS. If you have any questions, feel free to post them on CANVAS.